Anabel Yong

ay180501@gmail.com — Google Scholar: Link — Website: https://anabelyong.github.io/

Summary

I work at the intersection of probabilistic machine learning and chemical discovery, with a focus on uncertainty-aware, data-efficient design. My background in biochemistry and computer science, together with experience at IgnotaLabs.AI and recent work in Bayesian optimisation and LLMs, shaped my interest in molecular and materials discovery. I aim to build chemical tools that make machine learning more useful for chemists by supporting structured editing, optimisation, and confidence-aware decision-making. More specifically:

- **Bayesian decision-making** - Bayesian optimization and Pareto-aware strategies for chemical and materials design.
- **Uncertainty calibration** - confidence-aware models for safety- and cost-sensitive discovery.
- **LLMs for chemical reasoning** - language-based systems for structured editing, interpretation, and optimization in AI4Science settings.

Publications

- **Yong, A.**, Tripp, A., Hosseini-Gerami, L., Paige, B. (2025). A study of EHVI vs fixed scalarization for molecule design. (Accepted as Poster, NeurIPS 2025 AI for Science Workshop.) Link: arXiv

- Sun, H., **Yong, A.**, Gilly, L, Jin, F. (2025). Enhancing Instruction Following Capabilities in Seq2Seq Models: DoLA Adaptations for T5 (arXiv preprint) Link: arXiv

Education

MSc Computational Statistics & Machine Learning, UCL (Distinction) Sept 2023 - Sept 2024
Thesis Topic (Distinction): *"Multi-Objective Bayesian Optimization with Independent Tanimoto Kernel Gaussian Processes for Diverse Pareto Front Exploration"*
Modules: Probabilistic & Unsupervised Learning (Gatsby PhD module Coursework) (100%), Machine Learning Seminar(76%), AI for Biomedicine & Healthcare (70%), Statistical Natural Language Processing

BSc Biochemistry(with Mathematics), University of Edinburgh Oct 2019 - June 2023
Thesis Topic (1st Class Honours): *"Evaluation of Different False Discovery Rate approaches in Large-scale Proteomics Data"*
Modules: Bioinformatics (72%), Numerical Methods for Differential Equations (85%), Linear Programming, Modelling & Solutions (71%), Mathematics for Natural Sciences (80%), Biological Chemistry (85%)

Research Experience

- **Research Intern, National University of Singapore (PI: Jonathan Scarlett)** May 2025 - Sept 2025
  - Researched Bayesian optimization algorithms with diversity and safety constraints, focusing on uncertainty-aware batch selection for molecular design.
  - Benchmarked batch Bayesian optimization methods on multi-property optimization tasks, analyzing trade-offs between utility, Pareto front quality, and chemical diversity.

- **Machine Learning Research Intern, IgnotaLabs.AI** (**PIs: Dr. Austin Tripp, Cambridge MLG Group, Layla Hosseini-Gerami & Brooks Paige, UCL**) June 2024 - Sept 2024
  - Built exact Gaussian process models on molecular fingerprint features for data-efficient multi-objective molecular design.
  - Evaluated Pareto-aware Bayesian optimization against fixed-weight scalarization, with better recovery of chemically diverse trade-off solutions.

- **Research Intern, Kustatscher Lab, University of Edinburgh** Oct 2022 - July 2023
  - Benchmarked false discovery rate strategies for large-scale mass-spectrometry proteomics, improving confidence in protein and isoform identification.
  - Contributed to human protein database ProteomeHD2 curation by identifying previously uncharacterised novel microproteins, supporting expansion of the human proteome database.

- **Research Intern, ZiHeng Yang Lab, UCL** June 2022 – Sept 2022
  - Implemented Bayesian MCMC methods to model ABO blood-group frequencies, using probabilistic inference to study genetically determined molecular phenotypes in human populations.

Skills

Programming:	Python, R, MATLAB, git, Linux, Google Cloud Platform (GCP), slurm, LaTeX
Machine Learning:	PyTorch, Jax, BoTorch, GPyTorch
Biochemistry / Structural Biology:	Protein purification, SDS-PAGE, EMSA, mass-spectrometry proteomics, structural biology wet-lab methods
Bioinformatics / Cheminformatics:	ChemDraw, RDKit, PyMOL, UniProt, MaxQuant, AutoDock Vina, BLAST

Awards & Scholarships

- **Yayasan Sarawak Tun Taib Postgraduate MSc Scholarship 2023/2024 -** fully funded Master's / postgraduate scholarship, awarded to 1 of ≈830 applicants (value: ≈£50,000)
- **EPSRC Undergraduate Biochemistry Research Bursary, University of Edinburgh (2021) -** £2,500

Leadership & Service

- **Treasurer, CompSoc Edinburgh, University of Edinburgh** 2020-2021
- **Organising Team, HackTheBurgh VI -** helped organise Scotland's largest student hackathon. 2021
- **Organising Committee, ClimateHack.AI -** supported an international AI hackathon in climate change and sustainability with participating university teams from around the world. 2023

References (available upon request)

- **Brooks Paige, Associate Professor in Machine Learning, University College London:** b.paige@ucl.ac.uk
- **Jonathan Scarlett, Associate Professor in Dept. of Computer Science & Mathematics, National University of Singapore:** scarlett@comp.nus.edu.sg
- **Layla Hosseini-Gerami, Chief Data Science Officer, IgnotaLabs.ai:** layla.gerami@ignotalabs.ai
- **Austin Tripp, Senior Machine Learning Research Scientist, Valence Labs:** austin.james.tripp@gmail.com